PTF ALL

Air Conditioning Low Airflow From Air Conditioning Outlets - Fault Isolation

CORRECTIVE ACTION SUMMARY

Correct Related Maintenance Message
Run Conditioned Air for five minutes
Replace Left Pack Flow Control and Shutoff Valve [V184]
Do the applicable AMM Task(s)
Repair or replace the applicable Flight Deck Air Distribution Ducts
Replace Flight Deck Air Distribution System Muffler/Duct Assembly
Clean Duct for Flight Deck Air Distribution Ducts
Replace Left Standby Temperature Control Valve [V140]
Replace Right Standby Temperature Control Valve [V142]

A. Description

(1) Low airflow from the air conditioning outlets (airflow diffusers) may be caused by blockage in the outlets, disconnected or kinked flexible hoses/ducts, an excess accumulation of dust, lint, and other unwanted materials on the airflow restrictors (flow control orifices) installed between the air distribution ducts and flexible hoses, or by air distribution duct/hose joint separation.

(2) Low airflow from the air conditioning outlets causes a reduction in cooling capability that can lead to elevated temperatures in the airplane compartments and also cause overheating of electrical/ electronic equipment.

(3) Air leaking from a separated duct joint may also cause condensation to form and drip into the airplane interior. Duct joints that are suspectible to separation or leakage are usually located adjacent to the ECS fans or midway between long duct runs. Duct joints that are adjacent to ECS fans are subjected to higher air pressure, particularly when the air conditioning outlets are closed. Duct joints that are midway between long duct runs are subjected to flexing. Improperly installed metal clamps, plastic straps and flexible sleeves/hoses at the duct joints without sufficient overlap on the duct may not provide adequate clamping force to prevent separation of the duct joint.

B. Initial Evaluation

(1) Supply conditioned air to the airplane with the left pack. This is the task:
- Supply Conditioned Air with a Cooling Pack, AMM TASK 21-00-00-800-803.

(2) Do a check of the airflow from the flight deck air outlets.

(3) Remove conditioned air from the airplane. This is the task:
- Remove Conditioned Air Supplied by a Cooling Pack, AMM TASK 21-00-00-800-804.

(4) If the airflow was not sufficient, with conditioned air supplied, then do the Fault Isolation Procedure below.

(5) If the airflow was sufficient, with conditioned air supplied, then there was an intermittent fault.

C. Fault Isolation Procedure

(1) Look in OMF Existing Faults for this related maintenance message: 21-55030.

(a) If you find the related maintenance message, then do these steps:

1) Do the fault isolation task for the related maintenance message.

2) Supply conditioned air to the airplane with the left pack. This is the task:
- Supply Conditioned Air with a Cooling Pack, AMM TASK 21-00-00-800-803.

3) Do a check of the airflow from the flight deck air outlets.

4) Remove conditioned air from the airplane. This is the task:
- Remove Conditioned Air Supplied by a Cooling Pack, AMM TASK 21-00-00-800-804.

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5) If the airflow was sufficient, with conditioned air supplied, then you corrected the fault.

(b) If you do not find the related maintenance message, then continue.

(2) It is possible that the wire mesh screen assembly of the flight deck duct assembly has an accumulation of ice. This can occur when you run the air conditioning system for long periods of time with the airplane on the ground during certain weather conditions.

(a) Supply conditioned air to the airplane with the left pack. This is the task:
- Supply Conditioned Air with a Cooling Pack, AMM TASK 21-00-00-800-803.

(b) Set the CONT CAB temperature selector, P5-17S2 to the full W position.

(c) Wait a minimum of five minutes while the pack operates.

(d) Do a check of the airflow from the flight deck air outlets.

(e) Remove conditioned air from the airplane. This is the task:
- Remove Conditioned Air Supplied by a Cooling Pack, AMM TASK 21-00-00-800-804.

(f) If the airflow was sufficient, with conditioned air supplied, then you corrected the fault.

(g) If the airflow was not sufficient, with conditioned air supplied, then continue.

(3) Do an operational test of the left pack flow control and shutoff valve, V184. This is the task:
- Pack Flow Control Valve Operational Test, AMM TASK 21-51-00-700-802.

(a) If you find a problem during the test, then do these steps:

1) Replace the left pack flow control and shutoff valve, V184. These are the tasks:
- Left Flow Control and Shutoff Valve Removal, AMM TASK 21-51-01-000-801,
- Left Flow Control and Shutoff Valve Installation, AMM TASK 21-51-01-400-801.

2) Supply conditioned air to the airplane with the left pack. This is the task:
- Supply Conditioned Air with a Cooling Pack, AMM TASK 21-00-00-800-803.

3) Do a check of the airflow from the flight deck air outlets.

4) Remove conditioned air from the airplane. This is the task:
- Remove Conditioned Air Supplied by a Cooling Pack, AMM TASK 21-00-00-800-804.

5) If the airflow was sufficient, with conditioned air supplied, then you corrected the fault.

(b) If you do not find a problem during the test, then continue.

(4) For the flight deck duct assembly screen, do this task:
- Flight Deck Duct Assembly Screen Cleaning, AMM TASK 21-22-07-100-801.

(a) Supply conditioned air to the airplane with the left pack. This is the task:
- Supply Conditioned Air with a Cooling Pack, AMM TASK 21-00-00-800-803.

(b) Do a check of the airflow from the flight deck air outlets.

(c) Remove conditioned air from the airplane. This is the task:
- Remove Conditioned Air Supplied by a Cooling Pack, AMM TASK 21-00-00-800-804.

(d) If the airflow was sufficient, with conditioned air supplied, then you corrected the fault.

(e) If the airflow was not sufficient, with conditioned air supplied, then continue.

(5) Do this check of the flight deck air distribution ducts and the flight deck air distribution system muffler/duct assembly between the mix manifold outlet and the flight compartment outlets:

NOTE: The flight deck air distribution ducts are located on the left side of the forward cargo compartment and the EE bay. The flight deck air distribution system muffler/duct assembly is located in the mix bay.

(a) Supply conditioned air to the airplane with the left pack. This is the task:
- Supply Conditioned Air with a Cooling Pack, AMM TASK 21-00-00-800-803.

(b) Inspect the ducts for leakage, loose connections, or damage.

(c) Remove conditioned air from the airplane. This is the task:
- Remove Conditioned Air Supplied by a Cooling Pack, AMM TASK 21-00-00-800-804.

(d) If you found a problem during the inspection at the flight deck air distribution ducts, then do these steps:

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1) Repair or replace the applicable flight deck air distribution ducts.
2) Supply conditioned air to the airplane with the left pack. This is the task:
   - Supply Conditioned Air with a Cooling Pack, AMM TASK 21-00-00-800-803.
3) Do a check of the airflow from the flight deck air outlets.
4) Remove conditioned air from the airplane. This is the task:
   - Remove Conditioned Air Supplied by a Cooling Pack, AMM TASK 21-00-00-800-804.
5) If the airflow was sufficient, with conditioned air supplied, then you corrected the fault.

(e) If you found a problem during the inspection at the flight deck air distribution system muffler/duct assembly, then do these steps:
1) Replace the flight deck air distribution system muffler/duct assembly. These are the tasks:
   - Flight Deck Air Distribution System Muffler/Duct Assembly Removal, AMM TASK 21-22-05-000-801,
   - Flight Deck Air Distribution System Muffler/Duct Assembly Installation, AMM TASK 21-22-05-400-801.
2) Supply conditioned air to the airplane with the left pack. This is the task:
   - Supply Conditioned Air with a Cooling Pack, AMM TASK 21-00-00-800-803.
3) Do a check of the airflow from the flight deck air outlets.
4) Remove conditioned air from the airplane. This is the task:
   - Remove Conditioned Air Supplied by a Cooling Pack, AMM TASK 21-00-00-800-804.
5) If the airflow was sufficient, with conditioned air supplied, then you corrected the fault.

(f) If you did not find a problem during the inspection, then continue.

(6) To prepare for an inspection, remove the flight deck air distribution ducts.
(a) Inspect the duct for blockage.
(b) If you find blockage during the inspection, then do these steps:
1) Clean the blockage from the duct.
2) Re-install the flight deck air distribution ducts.
3) Supply conditioned air to the airplane with the left pack. This is the task:
   - Supply Conditioned Air with a Cooling Pack, AMM TASK 21-00-00-800-803.
4) Do a check of the airflow from the flight deck air outlets.
5) Remove conditioned air from the airplane. This is the task:
   - Remove Conditioned Air Supplied by a Cooling Pack, AMM TASK 21-00-00-800-804.
6) If the airflow was sufficient, with conditioned air supplied, then you corrected the fault.

(c) If you do not find blockage during the inspection, then continue.
(d) Re-install the flight deck air distribution ducts.

(7) To prepare for an inspection, remove the flight deck air distribution system muffler/duct assembly. This is the task:
- Flight Deck Air Distribution System Muffler/Duct Assembly Removal, AMM TASK 21-22-05-000-801.

(a) Inspect the flight deck air distribution system muffler/duct assembly for blockage.
(b) If you find debris during the inspection, then do these steps:
1) Install a new flight deck air distribution system muffler/duct assembly. This is the task:
   - Flight Deck Air Distribution System Muffler/Duct Assembly Installation, AMM TASK 21-22-05-400-801.
2) Supply conditioned air to the airplane with the left pack. This is the task:
   - Supply Conditioned Air with a Cooling Pack, AMM TASK 21-00-00-800-803.
3) Do a check of the airflow from the flight deck air outlets.
4) Remove conditioned air from the airplane. This is the task:
   - Remove Conditioned Air Supplied by a Cooling Pack, AMM TASK 21-00-00-800-804.

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5) If the airflow was sufficient, with conditioned air supplied, then you corrected the fault.

(c) If you do not find debris during the inspection, then continue.

(d) Re-install the flight deck air distribution system muffler/duct assembly. This is the task:
- Flight Deck Air Distribution System Muffler/Duct Assembly Installation, AMM TASK 21-22-05-400-801.

(8) Replace the left standby temperature control valve, V140. These are the tasks:
- Standby Temperature Control Valve Removal, AMM TASK 21-51-11-000-801,
- Standby Temperature Control Valve Installation, AMM TASK 21-51-11-400-801.

(a) Supply conditioned air to the airplane with the left pack. This is the task:
- Supply Conditioned Air with a Cooling Pack, AMM TASK 21-00-00-800-803.

(b) Do a check of the airflow from the flight deck air outlets.

(c) Remove conditioned air from the airplane. This is the task:
- Remove Conditioned Air Supplied by a Cooling Pack, AMM TASK 21-00-00-800-804.

(d) If the airflow was sufficient, with conditioned air supplied, then you corrected the fault.

(e) If the airflow was not sufficient, with conditioned air supplied, then continue.

(9) Replace the right standby temperature control valve, V142. These are the tasks:
- Standby Temperature Control Valve Removal, AMM TASK 21-51-11-000-801,
- Standby Temperature Control Valve Installation, AMM TASK 21-51-11-400-801.

(a) Supply conditioned air to the airplane with the left pack. This is the task:
- Supply Conditioned Air with a Cooling Pack, AMM TASK 21-00-00-800-803.

(b) Do a check of the airflow from the flight deck air outlets.

(c) Remove conditioned air from the airplane. This is the task:
- Remove Conditioned Air Supplied by a Cooling Pack, AMM TASK 21-00-00-800-804.

(d) If the airflow was sufficient, with conditioned air supplied, then you corrected the fault.

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